



Comparative analysis of parametric B-spline and Hermite cubic spline based methods for accurate ECG signal modeling

Alka Mishra^{*}, Surekha Bhusnur, Santosh Kumar Mishra, Pushpendra Singh

Bhilai Institute of Technology, Bhilai House, Durg, Chhattisgarh 491001, India

ARTICLE INFO

Keywords:

Electrocardiogram (ECG)
Parametric spline
B-spline
Hermite cubic spline
PRD
RMSE

ABSTRACT

Analyzing Electrocardiogram (ECG) signals is imperative for diagnosing cardiovascular diseases. However, evaluating ECG analysis techniques faces challenges due to noise and artifacts in actual signals. Machine learning for automatic diagnosis encounters data acquisition hurdles due to medical data privacy constraints. Addressing these issues, ECG modeling assumes a crucial role in biomedical and parametric spline-based methods have garnered significant attention for their ability to accurately represent the complex temporal dynamics of ECG signals. This study conducts a comparative analysis of two parametric spline-based methods—B-spline and Hermite cubic spline—for ECG modeling, aiming to identify the most effective approach for accurate and reliable ECG representation. The Hermite cubic spline serves as one of the most effective interpolation methods, while B-spline is an approximation method. The comparative analysis includes both qualitative and quantitative evaluations. Qualitative assessment involves visually inspecting the generated spline-based models, comparing their resemblance to the original ECG signals, and employing power spectrum analysis. Quantitative analysis incorporates metrics such as root mean square error (RMSE), Percentage Root Mean Square Difference (PRD) and cross correlation, offering a more objective measure of the model's performance. Preliminary results indicate promising capabilities for both spline-based methods in representing ECG signals. However, the analysis unveils specific strengths and weaknesses for each method. The B-spline method offers greater flexibility and smoothness, while the cubic spline method demonstrates superior waveform capturing abilities with the preservation of control points, a critical aspect in the medical field. Presented research provides valuable insights for researchers and practitioners in selecting the most appropriate method for their specific ECG modeling requirements. Adjustments to control points and parameterization enable the generation of diverse ECG waveforms, enhancing the versatility of this modeling technique. This approach has the potential to extend its utility to other medical signals, presenting a promising avenue for advancing biomedical research.

Introduction

Electrocardiography (ECG) stands as a pivotal diagnostic tool, meticulously documenting the intricate electrical activity unfolding within the heart. The ECG waveform consists of several characteristic features namely waves, segments and intervals that provide valuable information about the heart's rhythm, rate, and overall cardiac health [1]. By analyzing these features along with heart rate, rhythm, and other parameters, healthcare professionals can diagnose various cardiac conditions, including arrhythmias, myocardial infarction, electrolyte imbalances, and conduction abnormalities [2–4]. In the field of cardiology, Computer-Aided Diagnosis (CAD) systems have the potential to significantly improve diagnostic accuracy, aid in early detection of cardiac

abnormalities, and enhance patient care [5]. However, the presence of noise and artifacts in real-time biomedical signals, such as electrocardiograms (ECGs), poses challenges for accurate analysis [6]. Additionally, acquiring large volumes of real medical data for training and research purposes is often limited due to privacy concerns. To overcome these issues, artificial ECG generation techniques have emerged as a valuable tool [7].

Creating synthetic ECG signal that faithfully mirrors the distinctive characteristics of a genuine ECG signal poses a formidable challenge in the realm of biomedical signal processing. The ECG signal represents the electrical activity of the heart and contains important information about the heart's rhythm, rate, and various abnormalities. Creating a realistic artificial ECG signal involves capturing and reproducing the key features

^{*} Corresponding author.

E-mail address: alkatiwari23@gmail.com (A. Mishra).

<https://doi.org/10.1016/j.jelectrocard.2024.153783>

present in real ECG signals. These features include the P wave, QRS complex, and T wave, which correspond to specific electrical events in the heart's cardiac cycle. Additionally, the amplitude, duration, and morphology of these waveform components can vary depending on factors such as age, gender, and individual physiological differences [8–10]. Several approaches have been employed to generate artificial ECG signals. One common method is to use mathematical models that simulate the electrical behavior of the heart. These models typically incorporate mathematical equations representing the electrical activity in different regions of the heart and the propagation of the electrical signal throughout the cardiac tissue. By adjusting the model's parameters and simulating physiological conditions, realistic ECG signals can be generated [11–13]. In this context, approaches utilizing elementary trigonometric and linear functions, along with methods based on Gaussian pulse derivation, have been proposed for ECG signal generation. While these methods are straightforward, they come with limitations, such as the possibility of discontinuities, insufficient smooth transitions between signal segments, and challenges in adapting to various signal types [14–16]. Another approach involves the use of machine learning techniques. Researchers have developed algorithms that can learn from real ECG data and generate synthetic ECG signals based on the learned patterns. These algorithms can capture the statistical properties of real ECG signals, allowing them to produce realistic and diverse artificial signals but with drawbacks such as complexity, computational requirements, lack of interpretability, and susceptibility to overfitting [17–20].

Accurate modeling of electrocardiogram (ECG) signals plays a vital role in various biomedical applications, including cardiac diagnostics and monitoring. Parametric spline-based methods have gained significant attention due to their ability to capture the complex temporal dynamics of ECG waveforms. This study presents a comparative analysis of two parametric spline-based methods, namely, Hermite cubic spline and B-spline, for ECG modeling. The objective is to identify the strengths and limitations of each method and determine their suitability for accurate and reliable representation of ECG signals. These modeling techniques stand out in ECG signal representation due to their distinct advantages over other methods and offer a mathematical representation that aligns well with actual ECG signals, enabling comprehensive analysis and control over medical conditions without the need for extensive real data. B-Spline offers unparalleled flexibility with local control, allowing precise adjustments to individual control points without affecting the entire spline globally. This adaptability is crucial for accommodating diverse signal characteristics. Moreover, B-Spline ensures smooth transitions between segments, providing an aesthetically pleasing and continuous representation of the signal. Hermite cubic splines, on the other hand, excel in providing C2 continuity, ensuring continuous second derivatives and, consequently, a smooth and natural representation of the ECG waveform. Both methods are not only well-established and widely used but also exhibit mathematical elegance, making them suitable for theoretical analysis and practical applications in biomedical signal processing.

Methodology

This methodology outlines a systematic approach for the comparative analysis of B-spline and cubic spline methods in ECG modeling. It encompasses a mathematical understanding of the individual and shared features of these spline methods, detailing their implementation, and rigorous evaluation. B-spline and Hermite cubic spline were chosen due to their distinct advantages in representing the complex temporal dynamics of ECG signals. B-splines offer greater flexibility and smoothness, which are essential for approximating continuous signals with varying amplitudes and frequencies. Hermite cubic splines, on the other hand, are known for their superior interpolation capabilities, preserving critical control points that are vital for capturing the precise morphology of ECG waveforms. These properties make B-spline and

Hermite cubic spline particularly suitable for ECG modeling compared to other techniques such as linear interpolation, which may lack the necessary smoothness, or polynomial interpolation, which can suffer from oscillations.

B-spline and Hermite cubic spline methods have extensive applications in various fields such as computer-aided design and curve fitting, particularly when precise and smooth curve representations are paramount. Both methods are adept at generating smooth curves, making them suitable for scenarios where continuity and aesthetically pleasing shapes are crucial [21–23]. The utilization of parametric representations in these methods allows for meticulous control over the curves' shapes.

Considered continuity criteria

Zero Order Continuity (C_0): All data points of a wave are connected without breaks, ensuring continuity in terms of the data.

First Order Continuity (C_1): A curve exhibits first-order continuity (C_1) if it has a unique slope or a unique first derivative that is continuous at all data points.

Second Order Continuity (C_2): A wave with second-order continuity (C_2) possesses continuous second derivatives (curvature) at the points [22,23].

Parametric B-Spline method

The B-spline method is a mathematical technique for generating synthetic ECG signals that uses B-spline curves. B-splines are a type of piecewise-defined polynomial curves that can be used to approximate complex shapes with a series of control points. The curve is defined by $n + 1$ control points where n is highest subscript number of the control points, k is order of the polynomial and degree of the polynomial is $k - 1$. k also correlates to continuity of the curve and specifies, the number of control points that contribute in construction of a segment [24,25]. For example, if $k = 3$, then $k - 2 = 1$, i.e. curve will follow C_1 continuity (slope continuity) and every segment is influenced by 3 control points. The order k means that the curve is a piece wise polynomial segment of degree $k - 1$.

The parametric B-Spline function is given by

$$\mathbf{B}(\mathbf{u}) = \sum_{i=0}^{i=n} \mathbf{p}_i \mathbf{N}_{i,k}(\mathbf{u}), \quad (0 \leq \mathbf{u} \leq \mathbf{u}_{\max}) \quad (1)$$

where $\mathbf{p}_i$ are control points, $\mathbf{N}_{i,k}(\mathbf{u})$ are the normalized B-Spline basis functions, and $\mathbf{u}$ represents parameter (1).

The B-Spline basis function is expressed by the Cox-de boor recursion formula as (2)

$$\mathbf{N}_{i,k}(\mathbf{u}) = \frac{(\mathbf{u} - \mathbf{t}_i)\mathbf{N}_{i,k-1}(\mathbf{u})}{\mathbf{t}_{i+k-1} - \mathbf{t}_i} + \frac{(\mathbf{t}_{i+k} - \mathbf{u})\mathbf{N}_{i+1,k-1}(\mathbf{u})}{\mathbf{t}_{i+k} - \mathbf{t}_{i+1}} \quad (2)$$

where $\mathbf{t}_i$ ($0 \leq j \leq n + k$) are called knot vectors, it is also called a joint point at which a part of curve will join and the value is given by (3)

$$\begin{aligned} \mathbf{t}_j &= 0 \text{ if } i < k; \\ \mathbf{t}_j &= i - k + 1 \text{ if } k \leq i \leq n; \\ \mathbf{t}_j &= n - k + 2 \text{ if } i > n \\ \mathbf{N}_{i,k}(\mathbf{u}) &= 1 \text{ if } \mathbf{t}_j \leq \mathbf{u} \leq \mathbf{t}_{j+1} \text{ otherwise } 0 \end{aligned} \quad (3)$$

By applying these equations, a versatile array of synthetic ECG signals with desired morphology and characteristics can be produced [24]. The unique attributes and features of B-Spline are outlined below:

B-Spline Features:

- **Flexibility:** B-splines are highly flexible and allow for the representation of complex curves. The order of the B-spline can be adjusted to control the level of smoothness.

- Local Control: Changes to a control point affect only a local region of the curve, providing localized control. This feature allows for easier modifications without influencing the entire spline.
- Knot Vector: B-splines use a knot vector to define the parameterization of the curve. The placement of knots determines the influence of control points on the spline.
- Adaptability: B-splines are adaptable to changes in the number of control points without affecting the entire curve.
- Piecewise Representation: B-splines represent curves as a combination of polynomial segments, offering a piecewise construction.
- Smoothness: B-splines exhibit $C^{(k-1)}$ continuity, where 'k' is the order of the spline. Higher-order B-splines provide increased smoothness [24–26].

Parametric hermite cubic spline

Parametric Hermite cubic spline (or cubic spline for short) is a mathematical technique used for modeling and representing data points in a smooth and continuous manner, it is also called Hermite curve. It involves fitting a piecewise cubic polynomial function to a set of data points, where the function is defined by a set of control points and additional parameters [27]. The parametric cubic expression for a segment is represented by (1)

$$p(u) = \sum_{i=0}^3 C_i u^i, 0 \leq u \leq 1 \quad (4)$$

where, $p(u)$ is the position vector of a point on the curve is expressed as a function of u in parametric space, with C_i representing the polynomial coefficients. To determine the coefficients for a cubic Hermite curve, four boundary conditions are necessary, encompassing the first and last points on the curve. i.e. endpoints (p_0, p_1) and slope at the points i.e. (p'_0, p'_1) . By applying the boundary condition (p_0, p'_0) at $u = 0$ and (p_1, p'_1) at $u = 1$ for cubic spline curve at its endpoints (p_0, p_1) the final form of the Eq. (4) for cubic Hermite curve will be obtained as:

$$P(u) = (2u^3 - 3u^2 + 1)p_0 + (-2u^3 + 3u^2)p_1 + (u^3 - 2u^2 + u)p'_0 + (u^3 - u^2)p'_1, \quad 0 \leq u \leq 1 \quad (5)$$

The provided Eq. (5) illustrates a single cubic spline segment, which can be extended to generalize any two adjoining segments of a spline curve intended to fit a given number of data points. When dealing with a set of n data points, the cubic spline curve connects the endpoints with tangent vectors. However, for intermediate points, where tangent vectors are absent, achieving second-order continuity necessitates ensuring that the curvature remains continuous at these points. Connecting two Hermite spline curves to establish second-order continuity requires the equality of the second derivatives at the end of the first curve and the beginning of the second curve, thereby eliminating the need for additional tangent vectors.

$$p''(u_1 = 1) = p''(u_2 = 0) \quad (6)$$

By assigning the variable u to represent the segment number in the given relation, we can extend this relationship to calculate the tangent vector at the termination of the second curve. This vector, which also serves as the tangent vector at the commencement of the third curve, is derived by differentiating Eq. (5) and employing the result obtained from Eq. (6).

$$p'_1 = \frac{1}{4}(3p_0 + p'_0 - 3p_2 + p'_2) \quad (7)$$

The calculation of the unknown tangent vector or slope is achieved through Eq. (7). To extend this process for multiple segments, the same procedure is iteratively applied, resulting in the formulation of a matrix equation. Solving this matrix equation facilitates the determination of

intermediate tangent vectors, as highlighted in previous studies [27–29]. The distinctive characteristics and qualities of cubic splines are summarized as follows:

Cubic Spline Features:

- Interpolation: Cubic splines specifically use cubic polynomials to interpolate between data points.
- Global Interpolation: Changes to a control point have a global effect on the entire spline. Global nature ensures smooth transitions between segments.
- Endpoint Conditions: Cubic splines commonly use natural or clamped boundary conditions at the endpoints. Natural conditions set the second derivatives at endpoints to zero, ensuring smooth connections.
- Ease of Implementation: Cubic splines are relatively easy to implement and require solving a system of linear equations.
- Piecewise Cubics: Cubic splines represent the curve as a set of cubic polynomials between consecutive data points.
- Preservation of Control Points: Cubic splines inherently pass through each control point, preserving their positions on the curve [30].

Steps for generating the parametric B- Spline and Hermite curve as well as shared features

Common Steps.

1. Control Points: The first step is to define a set of control points that represent the desired shape and morphology of the ECG waveform. These control points typically include the peak of the P wave, the peak of the QRS complex, and the peak of the T wave.
2. Interpolation: Cubic spline interpolation is then applied to connect the control points with a smooth curve. The spline function consists of a series of cubic polynomials that smoothly interpolate between adjacent control points, ensuring continuity and smoothness of the resulting curve.
3. Parametric Equation: A parametric equation is used to describe the cubic spline curve. The parameter, typically denoted as u , ranges from 0 to 1 and represents the progression along the curve. By varying the parameter value, the ECG waveform can be generated at different points in time.
4. Parameterization: The parameter u can be uniformly or non-uniformly parameterized to control the timing of the ECG waveform. This allows for variations, in duration, and morphology of the ECG signal.

Specifically, for B-Spline

- Knot Vector: A knot vector is defined to determine the position and influence of each control point along the B-spline curve. The knot vector represents a sequence of values that determine the parameterization of the B-spline curve. It controls the shape and flexibility of the curve.
- B-spline Basis Functions: B-spline basis functions are mathematical functions that define the influence of each control point on the B-spline curve. These basis functions are defined recursively using the control points and knot vector. They determine how the curve is constructed by approximating the control points.
- Curve Generation: The B-spline curve is generated by combining the control points and basis functions. The curve smoothly approximates the control points, creating a smooth and continuous ECG waveform [28–30].

Simulation results

To simulate the ECG signal modeling process, an authentic ECG signal is extracted from the Physionet database [31]. Thirteen

strategically selected control points, representing a complete cycle of the ECG waveform [32], are chosen due to their medical significance. These control points serve as key markers, encompassing waves, segments, and intervals crucial for comprehensive cardiac assessment. These points were chosen based on their clinical relevance and their role in defining the morphology of the ECG signal [29]. Utilizing these control points, a parametric cubic spline and B-spline curve are crafted to faithfully depict the ECG waveform. Maintaining second-order continuity in both methods ensures smooth connections between data points, emphasizing slope and curvature consistency. This level of continuity proves sufficient for an accurate representation, fostering seamless connections for enhanced biomedical analysis and interpretation. The avoidance of higher-order polynomials mitigates computational complexities associated with potential issues.

In this section, the application of parametric B-Spline and parametric cubic spline in ECG modeling is demonstrated, focusing on Normal Sinus Rhythms and a cardiac disorder, namely Congestive Heart Failure. Figs. 1, 2, 3, and 4 showcase the capability of these models to accurately simulate key features associated with the mentioned medical conditions. Figs. 1 and 2 depict the modeling of both normal and abnormal ECG signals using the B-Spline method, juxtaposed with the corresponding original ECG signals. Similarly, Figs. 3 and 4 exhibit the modeling of normal and abnormal ECG signals using parametric cubic spline, alongside the real ECG signals. These visual representations illustrate the fidelity of the modeled ECG signals to the real ECG signals, offering a comparative analysis that visually assesses the performance of the modeling methods. As well as four quantitative measures have been considered to assess the quality of the modeled signal in comparison to the actual signal.

Performance measures

To obtain a comprehensive understanding of the similarity or dissimilarity between the original ECG and modeled ECG signals, following quantitative measures were employed:

- **Power Spectrum Analysis:** This analysis involves decomposing a signal into its frequency components. It helps identify the frequency composition of a signal, highlighting dominant frequencies and their contributions [33,34]. This analysis is relevant to ECG modeling as it helps in identifying any discrepancies in the frequency domain, which could indicate potential distortions or loss of critical information in the modeled signals. This power spectrum analysis is

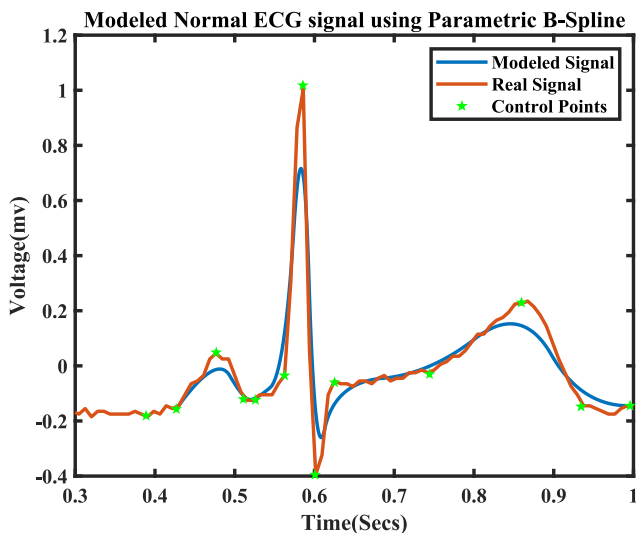


Fig. 1. Modeled Normal ECG signal using the B-Spline approach along with the original signal.

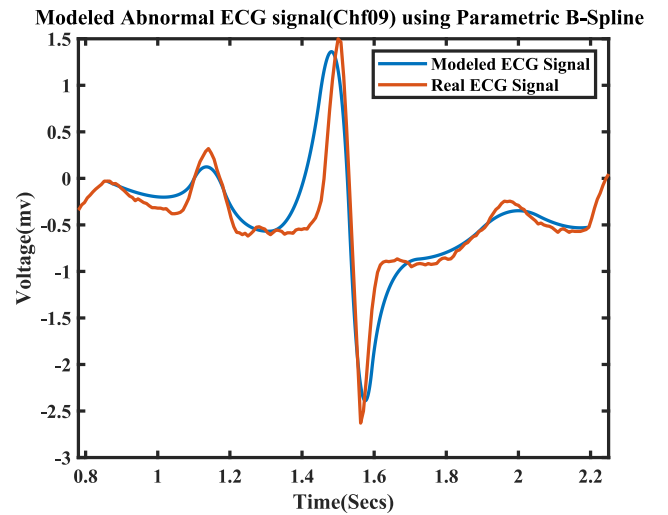


Fig. 2. Modeled Abnormal ECG signal using the B-Spline approach along with the original signal.

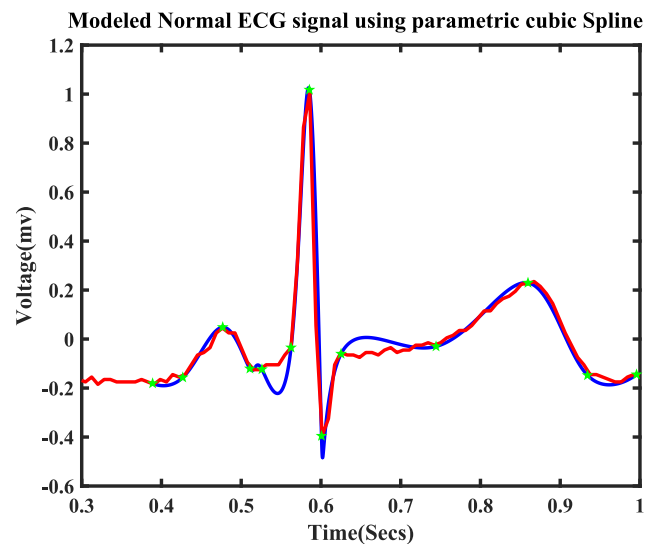


Fig. 3. Modeled Normal ECG signal using the Cubic spline approach along with the original signal.

employed for both B-Spline modeling and cubic spline modeling, as illustrated in Fig. 5 and Fig. 6, respectively, that can reveal the strength or intensity of different frequency components present in the signals.

- **Cross-Correlation:** It measures the similarity between two signals as a function of a time lag applied to one of them. It indicates how much one signal resembles another when one is shifted over time. In the realm of ECG analysis, cross-correlation serves as a valuable tool to evaluate the alignment and similarity between the original and modeled ECG signals [35,36]. Fig. 7 presents the real ECG signal alongside the Aligned synthetic ECG signal generated through the B-Spline method. Similarly, Fig. 8 depicts the Real ECG signal in conjunction with the Aligned synthetic ECG signal produced by the Cubic Spline method. Notably, both alignments exhibit a time lag of 0.000 s. The correlation coefficient for the Cubic Spline method is measured at 0.977, while the B-Spline method demonstrates a correlation coefficient of 0.953.
- **Root Mean Square Error (RMSE):** It measures the average magnitude of the differences between corresponding values of two signals. It quantifies the overall accuracy of a model or estimation by

Modeled Abnormal ECG signal(Chf09) using parametric cubic Spline

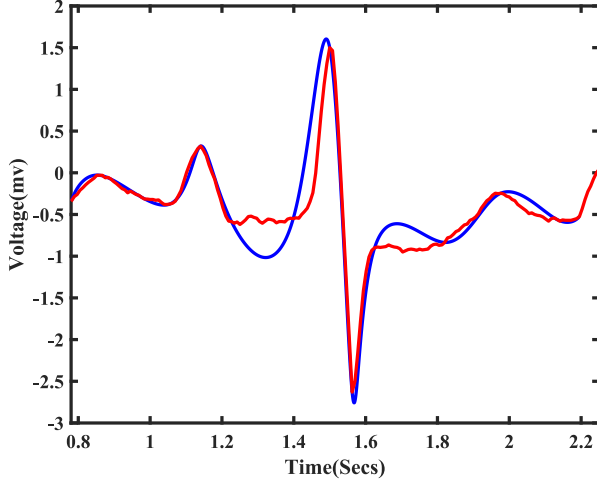


Fig. 4. Modeled Abnormal ECG signal using the Cubic approach along with the original signal.

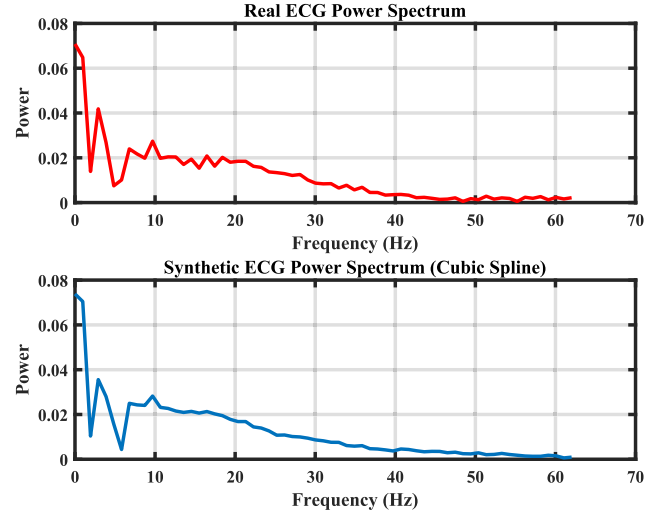


Fig. 6. Power Spectrum analysis of synthetic ECG (Cubic Spline) along with the original ECG signal.

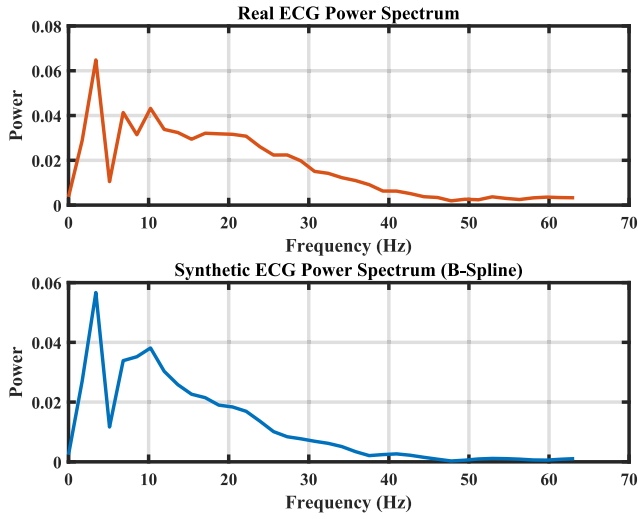


Fig. 5. Power Spectrum analysis of synthetic ECG (B-Spline) along with the original ECG signal.

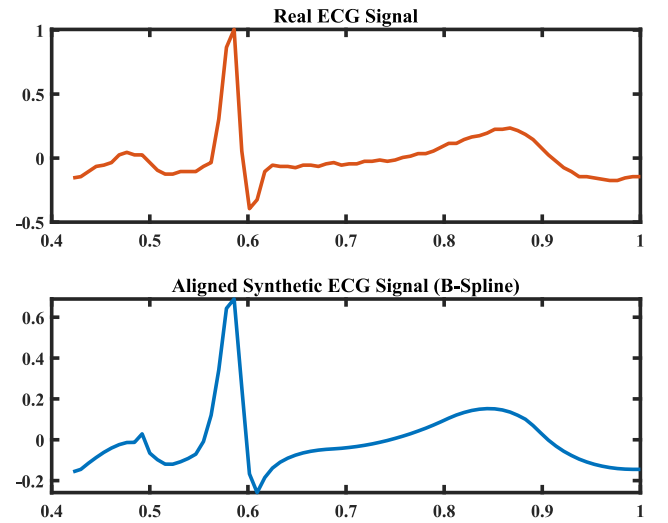


Fig. 7. Real ECG signal along with Aligned synthetic ECG signal (B-Spline).

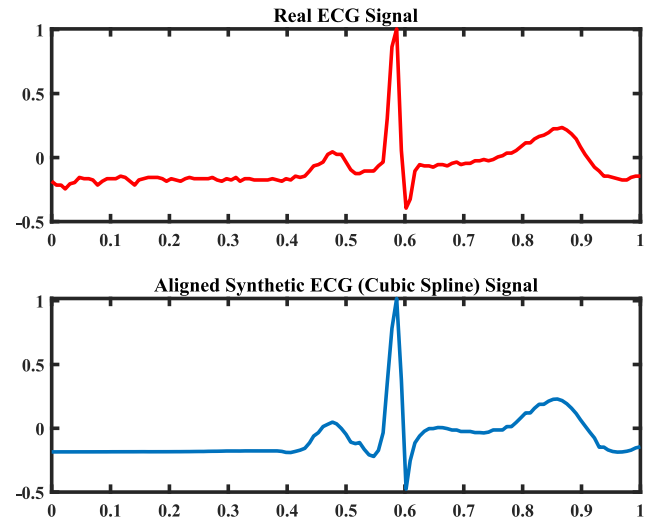


Fig. 8. Real ECG signal along with Aligned synthetic ECG signal (Cubic Spline).

calculating the square root of the average squared differences [37]. In ECG analysis, RMSE provides a numerical measure of the discrepancy between the original and modeled signals expressed in (8).

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2} \quad (8)$$

➤ **Percentage Root Mean Square Difference (PRD):** It is a normalized version of RMSE, expressed as a percentage. It represents the RMSE relative to the magnitude of the original signal [38]. PRD is often used to compare signals with different scales, providing a percentage measure of the average discrepancy between the original and modeled signals in (9)

$$\text{PRD} = \sqrt{\frac{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2}{\sum_{i=1}^n (y_i)^2}} \times 100 \quad (9)$$

These performance parameters provide numerical values that

Table 1

RMSE and PRD values for both normal and abnormal ECG signal taking into account B-Spline and cubic modeling.

Signal	RMSE (Cubic Spline modeling)	PRD (Cubic Spline modeling)	RMSE (B-Spline modeling)	PRD (B-Spline modeling)
Normal Beat_1	0.13278	38.726	0.14774	42.557
Normal Beat_2	0.07952	36.803	0.15334	44.003
Normal Beat_3	0.07769	36.020	0.1688	45.657
Normal Beat_4	0.09752	38.083	0.1793	46.433
Abnormal Beat_1	0.6443	35.023	0.1411	42.075
Abnormal Beat_2	0.0889	37.954	0.1592	44.125
Abnormal Beat_3	0.07632	36.040	0.1643	45.023
Abnormal Beat_4	0.0687	35.988	0.14932	43.466

indicate the accuracy of the modeled ECG signal. Lower values of PRD and RMSE indicate a closer match between the actual and modeled signals, suggesting a more accurate modeling process. Where n is total number of data points considered for calculation, $\hat{y}$ and y_i represent modeled signal actual signal respectively. The Table 1 displays the RMSE and PRD values for both four normal and four abnormal ECG conditions (Congestive Heart Failure), taking into account B-Spline and cubic modeling.

Discussion

In both the parametric B-spline and parametric cubic spline methods, cubic polynomial functions skilfully shape spline curves, ensuring seamless transitions and consistency between segments. This meticulous process is executed using MATLAB 2021 software. Parametric B-splines are employed to approximate the ECG signal, constructing piecewise polynomial functions that closely align with the given data points. The manipulation of control points and the knot vector is undertaken to achieve an accurate representation of the ECG waveform.

Upon a comprehensive evaluation, considering visual comparisons and quantitative measures such as power spectrum analysis, cross-correlation, RMSE, and PRD, with one normal real ECG signal, it becomes evident that the parametric cubic spline method outperforms the B-spline method. The correlation coefficient for the Cubic Spline method, standing at 0.977, surpasses the correlation coefficient of the B-Spline method, which is measured at 0.953. Examining the maximum PRD and RMSE values, B-Spline records 46.433 and 0.1793, respectively, while cubic spline showcases maximum PRD and RMSE values of 38.726 and 0.13278. Examination of PRD and RMSE values revealed the robustness of the cubic spline in faithfully representing the original data. This discrepancy can be attributed to the inherent nature of the parametric cubic spline as an interpolation method, mandating the curve to pass through all control points, thereby ensuring a closer fit to the original data. On the contrary, B-spline, functioning as an approximation method, does not necessarily traverse all control points, resulting in a slightly less precise representation of the data.

Both approaches, Cubic Spline and B-Spline, showcase distinct characteristics and advantages. The Cubic Spline method, with its global interpolation capabilities and inherent control point preservation, excels in maintaining smooth transitions and accurate representation of the given data. On the other hand, the B-Spline method offers local control over the shape of the curve, flexibility, adaptability, and a piecewise construction approach, making it suitable for applications requiring nuanced adjustments and modular representation. The evaluation of the generated artificial ECG signals is crucial. The validation process often involves comparing statistical measures, waveform morphologies, and clinical features of the artificial signals with those of real ECG data. By advancing our understanding of the underlying physiology and employing sophisticated signal processing techniques, researchers continue to make progress in the generation of artificial ECG signals that closely resemble real-world recordings.

The flexibility of presented spline-based models also allows for the simulation of various cardiac conditions, which can be useful in training medical professionals and developing new diagnostic tools.

Additionally, these models can be employed in telemedicine to improve remote monitoring of patients with cardiac conditions, ensuring timely medical interventions. The principles of this research can be extended to model other biomedical signals such as electroencephalograms (EEGs) and electromyograms (EMGs), thereby broadening its impact across different areas of biomedical research.

Conclusion

This comparative analysis of parametric spline-based methods—specifically B-spline and cubic spline—provides valuable insights into their efficacy for accurate ECG signal modeling. The preliminary results indicate promising capabilities for both B-spline and cubic spline methods in representing ECG signals. The B-spline method stands out for its greater flexibility and smoothness, making it a suitable choice for applications requiring adaptability. On the other hand, the cubic spline method excels in capturing waveforms with precision, emphasizing the importance of control point preservation in the medical field. In this presented analysis, considered measures collectively offer insights into the accuracy, alignment, and overall similarity and difference between the original and modeled signals, aiding in the evaluation of the modeling technique's performance. As a promising tool in biomedical signal processing, the comparative analysis contributes to the ongoing efforts to improve ECG modeling accuracy, furthering our understanding of cardiac activity and supporting enhanced clinical diagnostics and treatment. The findings pave the way for continued exploration and refinement of parametric spline-based methods, with potential applications extending beyond ECG signals to other areas of medical research and signal processing.

Ethics approval

This research does not require ethics approval.

Funding

Not applicable.

CRediT authorship contribution statement

Alka Mishra: Writing – original draft, Visualization, Validation, Methodology, Data curation, Conceptualization. **Surekha Bhusnur:** Writing – review & editing, Visualization, Supervision. **Santosh Kumar Mishra:** Visualization, Validation, Software, Methodology. **Pushpendra Singh:** Writing – review & editing, Visualization, Software, Resources.

Declaration of competing interest

Authors have no competing interests.

References

- [1] Chaves L, Marques G. Data mining techniques for early diagnosis of diabetes: a comparative study. *Appl Sci* 2021;11(5):2218.

- [2] Chatterjee S, Thakur RS, Yadav RN, Gupta L, Raghuvanshi DK. Review of noise removal techniques in ECG signals. *IET Signal Process* 2020;14(9):569–90.
- [3] Onak ON, Dogrusoz YS, Weber GW. Robustness of reduced order non-parametric model for inverse ECG solution against modeling and measurement noise. In: 2018 computing in cardiology conference (CinC). vol. 45. IEEE; 2018, September. p. 1–4.
- [4] Agrawal AK, Mishra A, Kumar R. ECG signal compression using efficient transformations. *Int J Innov Eng Technol* 2014;4(1):13–9.
- [5] Yanase J, Triantaphyllou E. A systematic survey of computer-aided diagnosis in medicine: past and present developments. *Expert Syst Appl* 2019;138:112821.
- [6] Tripathi PM, Kumar A, Komaragiri R, Kumar M. A review on computational methods for denoising and detecting ECG signals to detect cardiovascular diseases. *Arch Comput Meth Eng* 2021;1–40.
- [7] Mishra A, Bhusnur S. A new adaptive modeling and Denoising of real ECG signal. In: 2022 IEEE 3rd global conference for advancement in technology (GCAT). IEEE; 2022 October. p. 1–6.
- [8] Roopa CK, Harish BS. A survey on various machine learning approaches for ECG analysis. *Int J Comp Appl* 2017;163(9):25–33.
- [9] Sahoo S, Dash M, Behera S, Sabut S. Machine learning approach to detect cardiac arrhythmias in ECG signals: a survey. *Irbm* 2020;41(4):185–94.
- [10] Mishra A, Bhusnur S. Recapitulation of synthetic ECG signal generation methods and analysis. *Int J Signal Processing Systems* 2022;10(3):14–7.
- [11] Mishra A, Bhusnur S, Mishra SK. Advancing health sciences and biomedical technology: a parametric spline based ECG modeling. *Everyman's Sci* 2022;57(2): 1–5.
- [12] Boulakia M, Cazeau S, Fernández MA, Gerbeau JF, Zemzemi N. Mathematical modeling of electrocardiograms: a numerical study. *Ann Biomed Eng* 2010;38: 1071–97.
- [13] Niederer SA, Campbell KS, Campbell SG. A short history of the development of mathematical models of cardiac mechanics. *J Mol Cell Cardiol* 2019;127:11–9.
- [14] Dolinský P, András I, Michaeli L, Saliga J. An ECG signal model based on a parametric description of the characteristic waves. *Acta IMEKO* 2020;9(2).
- [15] Talbi ML, Ravier P. Flexible ECG signal modeling and compression using alpha stable functions. *Med Eng Phys* 2022;109:103865.
- [16] Gerasimov AK, Pedonova ZN. Development of hardware and software for generating test ECG signals. *Biomed Eng* 2022;55(5):315–9.
- [17] Adib E, Afghah F, Prevost JJ. Synthetic ECG signal generation using generative neural networks. *arXiv preprint* 2021. arXiv:2112.03268.
- [18] Zhu F, Ye F, Fu Y, Liu Q, Shen B. Electrocardiogram generation with a bidirectional LSTM-CNN generative adversarial network. *Sci Rep* 2019;9(1):6734.
- [19] Hazra D, Byun YC. SynSigGAN: generative adversarial networks for synthetic biomedical signal generation. *Biology* 2020;9(12):441.
- [20] Wasimuddin M, Elleithy K, Abuzneid AS, Faezipour M, Abuzagheh O. Stages-based ECG signal analysis from traditional signal processing to machine learning approaches: a survey. *IEEE Access* 2020;8:177782–803.
- [21] Kalkan E, Okur FY, Altunışık AC. Applications and usability of parametric modeling. *J Construct Eng Manag Innov* 2018;1(3):139–46.
- [22] Li J, Liu C. A quartic trigonometric interpolatory spline with local free parameters. *Adv Mech Eng* 2023;15(6). p.16878132231180154.
- [23] Herzog R, Blanc P. Optimal G2 Hermite interpolation for 3D curves. *Comp Aided Des* 2019;117:102752.
- [24] Liu D, Gu D, Smyl D, Deng J, Du J. B-spline level set method for shape reconstruction in electrical impedance tomography. *IEEE Trans Med Imaging* 2019; 39(6):1917–29.
- [25] Liu D, Gu D, Smyl D, Deng J, Du J. Multiphase conductivity imaging with electrical impedance tomography and B-spline level set method. *IEEE Trans Instrumentation Measurement* 2020;69(12):9634–44.
- [26] Mishra A, Bhusnur S, Mishra SK, Singh P. Exploring a new frontier in cardiac diagnosis: ECG analysis enhanced by machine learning and parametric quartic spline modeling. *J Electrocardiol* 2024;85:19–24.
- [27] Jafari A, Pszczolkowski E, Krishnamurthy A. A framework for biomechanics simulations using four-chamber cardiac models. *J Biomech* 2019;91:92–101.
- [28] Mishra A, Bhusnur S. A piecewise spline approach for modeling of ECG signals. *Biomed Phys Eng Express* 2023;9(6):065017.
- [29] Mishra A, Bhusnur S, Mishra SK, Singh P. Innovative ECG signal modeling: unleashing the potential of the parametric spline approach for advancing Global Health challenges. *J Tech Educ* 2023:160.
- [30] Zeid I. CAD/CAM theory and practice. McGraw-Hill Higher Education; 1991.
- [31] Goldberger AL, Amaral LA, Glass L, Hausdorff JM, Ivanov PC, Mark RG, et al. PhysioBank, PhysioToolkit, and PhysioNet: components of a new research resource for complex physiologic signals. *Circulation* 2000;101(23):e215–20.
- [32] Mishra A, Bhusnur S, Mishra S. A neoteric parametric representation and classification of ECG signal. *Circuits Syst Signal Process* 2023:1–14.
- [33] Chiani M, Elzanaty A. On the LoRa modulation for IoT: waveform properties and spectral analysis. *IEEE Internet Things J* 2019;6(5):8463–70.
- [34] Lee D, Baldick R. Future wind power scenario synthesis through power spectral density analysis. *IEEE Trans Smart Grid* 2013;5(1):490–500.
- [35] Bachura M, Fischer T. Waveform cross-correlation for differential time measurement: Bias and limitations. *Seismol Res Lett* 2019;90(5):2005–14.
- [36] Zang X, Brown MG, Godin OA. Waveform modeling and inversion of ambient noise cross-correlation functions in a coastal ocean environment. *J Acoust Soc Am* 2015; 138(3):1325–33.
- [37] Chiang HT, Hsieh YY, Fu SW, Hung KH, Tsao Y, Chien SY. Noise reduction in ECG signals using fully convolutional denoising autoencoders. *Ieee Access* 2019;7: 60806–13.
- [38] Bing P, Liu W, Zhang Z. DeepCEDNet: an efficient deep convolutional encoder-decoder networks for ECG signal enhancement. *IEEE Access* 2021;9:56699–708.